

Literature-Implied Status Note for arXiv:1806.09366

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Status

literature_implied_answer (partial subcase).

This note records a known-theorem implication for one scalar-coordinate reading of the reciprocal Bishop–Phelps–Bollobas question in Choi–Dantas–Jung–Martin, *The Bishop-Phelps-Bollobas property and absolute sums*, arXiv:1806.09366.

Source Question

In Section 3, after proving inheritance results for absolute summands in the domain, the source paper discusses reciprocal results. It observes that although each pair $(\mathbb{R}, Y)$ has the BPBp trivially, there are Banach spaces Y such that $(\mathbb{R} \oplus_1 \mathbb{R}, Y)$ does not have the BPBp; since $\mathbb{R} \oplus_1 \mathbb{R}$ is linearly isometric to $\mathbb{R} \oplus_\infty \mathbb{R}$, the same example also treats the ℓ_∞ scalar sum. The source then asks what happens for ℓ_p -sums with $1 < p < \infty$.

Supporting Literature

Kim–Lee, *Uniform convexity and the Bishop-Phelps-Bollobas property*, Canad. J. Math. 66 (2014), Theorem 3.1, proves that if the domain space X is uniformly convex, then (X, Y) has the BPBp for every Banach space Y . The same theorem is quoted in Dantas–Kadets–Kim–Lee–Martin, arXiv:1709.00032, Introduction, in the form: among known BPBp results, (X, Y) has the BPBp when X is uniformly convex, citing Kim–Lee Theorem 3.1 and Acosta–Becerra Guerrero–Meroño Theorem 2.2.

Identification

For every $1 < p < \infty$, the two-dimensional space

$$\mathbb{R} \oplus_p \mathbb{R} = \ell_p^2$$

is uniformly convex. Therefore Kim–Lee’s uniformly convex-domain theorem applies directly and gives that

$$(\ell_p^2, Y)$$

has the BPBp for every Banach space Y .

Consequently, the specific scalar-coordinate obstruction used in the source for $p = 1$ and $p = \infty$ cannot occur for $1 < p < \infty$: even though the coordinate pairs $(\mathbb{R}, Y)$ are only trivially BPBp, the full scalar ℓ_p^2 domain is automatically BPBp with every range.

Scope

This is not a new proof and not a full solution to the arbitrary reciprocal problem for $X_1 \oplus_p X_2$. It answers only the scalar two-summand subcase $X_1 = X_2 = \mathbb{R}$ that is naturally suggested by the source paper's displayed $\mathbb{R} \oplus_1 \mathbb{R}$ and $\mathbb{R} \oplus_\infty \mathbb{R}$ counterexample discussion. The general implication

$$(X_1, Y), (X_2, Y) \text{ have BPBp} \implies (X_1 \oplus_p X_2, Y) \text{ has BPBp}$$

for arbitrary Banach spaces remains outside the scope of this packet.

Provenance

Local evidence used:

- arXiv:1806.09366 source, Section 3, line 451 in the parsed TeX.
- arXiv:1709.00032 source, Introduction, quoting the uniformly convex-domain theorem and the ℓ_1^2 obstruction context.
- arXiv:1605.00245 source, Example after Proposition 3.??, quoting Kim–Lee Corollary 3.3 for the two-dimensional real characterization.

References

- [1] Y. S. Choi, S. Dantas, M. Jung, and M. Martin, *The Bishop-Phelps-Bollobas property and absolute sums*, arXiv:1806.09366.
- [2] S. K. Kim and H. J. Lee, *Uniform convexity and the Bishop-Phelps-Bollobas property*, Canad. J. Math. **66** (2014), 373–386.
- [3] S. Dantas, V. Kadets, S. K. Kim, H. J. Lee, and M. Martin, *There is no operatorwise version of the Bishop-Phelps-Bollobas property*, arXiv:1709.00032.